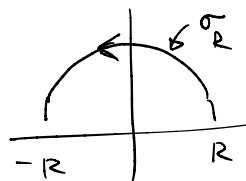


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Asmar & Grotztorf, Cor. 5.44, p. 324

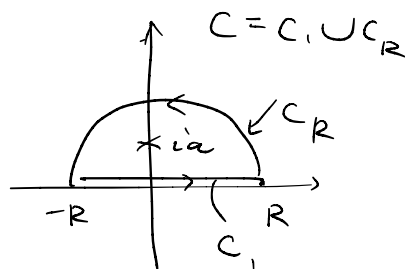
Cor Let p, q be complex polynomials w/
 $\deg(q) \geq 1 + \deg(p)$. Let σ_R denote the semicircular
 arc consisting of all $z = Re^{i\theta}$, w/ $0 \leq \theta \leq \pi$, then

$$\lim_{R \rightarrow \infty} \int_{\sigma_R} e^{isz} \frac{p(z)}{q(z)} dz = 0 \quad \text{if } s > 0$$



Ex 5.4.5 (p. 324) $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx, \quad a > 0$

$$\oint_C \frac{e^{iz} z}{z^2 + a^2} dz$$



$$= \underbrace{\int_{C_1} \frac{e^{iz} z}{z^2 + a^2} dz}_{\rightarrow 0 \text{ as } R \rightarrow \infty} + \underbrace{\int_{C_R} \frac{e^{iz} z}{z^2 + a^2} dz}_{\rightarrow 0 \text{ as } R \rightarrow \infty}$$

$$= \int_{-R}^R \frac{e^{ix} x}{x^2 + a^2} dx$$

$$\xrightarrow{R \rightarrow \infty} \int_{-\infty}^{\infty} \frac{x \cos x}{x^2 + a^2} dx + i \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx$$

$$\oint_C \frac{e^{iz} z}{z^2 + a^2} dz = 2\pi i \operatorname{Res}\left(\frac{e^{iz} z}{z^2 + a^2}, ia\right)$$

$$= 2\pi i \left. \frac{e^{iz} z}{\frac{d}{dz}(z^2+a^2)} \right|_{z=ia} = 2\pi i \frac{e^{i(ia)}(ia)}{2ia}$$

$$= i\pi e^{-a}$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{x \sin x}{x^2+a^2} = \operatorname{Im} \left(\oint_C \frac{e^{iz} z}{z^2+a^2} dz \right) = \pi e^{-a}$$

$$\underline{Ex} \quad \sum_{k=-\infty}^{\infty} \frac{1}{k^2+1} = ?$$

$$\text{Convention} \quad \sum_{k=-\infty}^{\infty} = \sum_{k \in \mathbb{Z}} = \lim_{N \rightarrow \infty} \sum_{k=-N}^N$$

Prop 5.6.2 (p. 347)

Suppose $f = \frac{p}{q}$ is a rational function w/ $\deg(p) \geq 2 + \deg(q)$

Suppose further that f has no poles at the integers.

Then

$$\sum_{k=-\infty}^{\infty} f(k) = -\pi \sum_j \operatorname{Res}(f(z) \cot(\pi z), z_j)$$

where the (finite) sum on the right runs over all the poles z_j of f .

$$\text{Ex } \sum_{k=-\infty}^{\infty} \frac{1}{k^2 + 1}$$

Consider $f(z) = \frac{1}{1+z^2}$, then

$$\sum_{k=-\infty}^{\infty} \frac{1}{k^2 + 1} = -\pi \left(\text{Res}\left(\frac{1}{1+z^2} \cot(\pi z), i\right) + \text{Res}\left(\frac{1}{1+z^2} \cot(\pi z), -i\right) \right)$$

$$\text{Res}\left(\frac{1}{1+z^2} \cot(\pi z), i\right) = \frac{\cot(\pi z)}{\frac{d}{dz}(1+z^2)} \Big|_{z=i} = \frac{\cot(\pi i)}{2i}$$

$$= \frac{\cos(\pi i)}{2i \sin(\pi i)} = \frac{\cosh(\pi)}{2i \cdot i \sinh(\pi)} = -\frac{1}{2} \coth(\pi)$$

$$\text{Res}\left(\frac{1}{1+z^2} \cot(\pi z), -i\right) = \frac{\cot(\pi z)}{\frac{d}{dz}(1+z^2)} \Big|_{z=-i} = \frac{\cot(-\pi i)}{-2i}$$

$$= \frac{\cos(-\pi i)}{-2i \sin(-\pi i)} = \frac{\cos(\pi i)}{2i \sin(\pi i)} = \dots = -\frac{1}{2} \coth(\pi)$$

$$\left(\sin(iy) = i \sinh(y) \text{ \& } \cos(iy) = \cosh(y) \right)$$

$$\Rightarrow \sum_{k=-\infty}^{\infty} \frac{1}{k^2 + 1} = -\pi \left(-\frac{1}{2} \coth(\pi) - \frac{1}{2} \coth(\pi) \right) = \pi \coth(\pi)$$

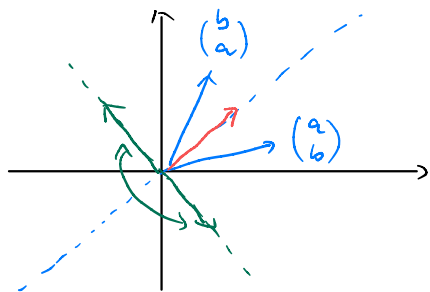
check p.348 for other examples.

Eigenvalues & Eigenvectors

- $Ax = b$ (solving linear eqns)
- $Ax = \lambda x$ (eigenvalue problem)

Ex $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$A: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} b \\ a \end{pmatrix}$$



e.g. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = +1 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

e.g. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Def Given a matrix $A \in M_n(\mathbb{R})$, If $\exists v \in \mathbb{R}^n \setminus \{0\}$ and $\lambda \in \mathbb{C}$ s.t.

$$Av = \lambda v$$

then λ is an eigenvalue of A

and v is an eigenvector of A

associated w/ λ ,

Rem • An eigenvalue could be 0 (scalar)

- An eigenvector cannot be 0 (vector)

$$\underline{\text{Ex}} \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Ax = \lambda x, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \neq 0$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{pmatrix} \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\neq 0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \ker \begin{pmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{pmatrix} \neq \{0\}$$

$$\Rightarrow \det \begin{pmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{pmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 = 0 \Rightarrow \lambda = 1, 1$$

$$\text{Let } \lambda = 1 \Rightarrow \begin{pmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\text{so we solve } \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (\text{solve } \ker \begin{pmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{pmatrix})$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}$$

In general, for $A \in M_n(\mathbb{R})$, the eigenvalues of A is given by the roots of the polynomial eqn

$$\det(A - \lambda I_n) = 0$$

characteristic
polynomial

$\hookrightarrow$ n -th-order polynomial in λ

The associated eigenvectors v to the eigenvalue λ_0 is given by solving the linear eqn

$$(A - \lambda_0 I_n) v = 0$$

(I_n is the $n \times n$ identity matrix)

FACT Any matrix $A \in M_n(\mathbb{R})$ has at least 1 eigenvector.

Ex $A = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

$$\begin{aligned} \det(A - \lambda I_2) &= \det \begin{pmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{pmatrix} \\ &= \lambda^2 - 6\lambda + 5 = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 5 \end{aligned}$$

For 2×2 matrices, the characteristic polynomial is given by

$$p(\lambda) = \det(A - \lambda I_2) = \lambda^2 - (\text{tr} A)\lambda + \det(A) \quad \underline{\hspace{1cm}}$$

For $\lambda_1 = 1$, Let's solve

$$\begin{pmatrix} 4-1 & 1 \\ 3 & 2-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ -3 \end{pmatrix} \quad \alpha \in \mathbb{R}$$

$$\Rightarrow v_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ is an e-vec for the e-val } \lambda_1 = 1$$

For $\lambda_2 = 5$, solve

$$\begin{pmatrix} 4-5 & 1 \\ 3 & 2-5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 & 1 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \alpha \in \mathbb{R}$$

$\Rightarrow v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an e-vec for the e-val $\lambda_2 = 5$

$$A v_1 = 1 v_1$$

$$A v_2 = 5 v_2$$

$$A \begin{bmatrix} \overset{\curvearrowright}{v_1} & v_2 \end{bmatrix} \quad \left(= \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} A v_1 & A v_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 v_1 & 5 v_2 \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\Rightarrow A \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{-1} \quad \text{diagonal}$$

e.g., $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix}^{-1}$

$$A = P \Lambda P^{-1}$$

(P invertible)
(Λ diagonal)

This is called a diagonalization of A

& A is called diagonalizable.

$$A^n = \underbrace{(P \Lambda P^{-1}) (P \Lambda P^{-1}) \dots (P \Lambda P^{-1})}_{n \text{ terms}}$$

$$= P \underbrace{\Lambda \dots \Lambda}_{n \text{ terms}} P^{-1} = P \Lambda^n P^{-1}$$

$$\Lambda^n = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} = \dots = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}$$

$$\Rightarrow A^n = P \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} P^{-1}$$

$$\Rightarrow \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^n = \begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix}^{-1}$$

$$e^{tA} = I_2 + (tA) + \frac{(tA)^2}{2!} + \frac{(tA)^3}{3!} + \dots$$

$$\Rightarrow e^{tA} = P e^{t\Lambda} P^{-1}$$

$$\text{w/ } e^{t\Lambda} = \begin{pmatrix} e^{t\lambda_1} & 0 \\ 0 & e^{t\lambda_2} \end{pmatrix}$$